Psychological processes of graphical perception: Past findings and our research proposal

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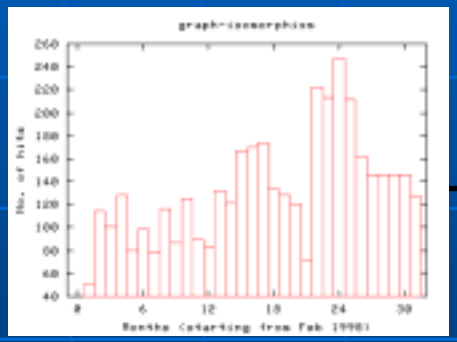
Outline

- **A model of graphical information-processing**
  1. Input: quantitative/qualitative information
  2. Retinal image
  3. Visual cortex
  4. Short-term and long-term memory
- **Implications**
- **Research proposal**
  1. Variables in the study
  2. Hypotheses
  3. Sample
  4. Materials
  5. Procedure
- **References**

A model of graphical information-processing (Kosslyn, Wilkinson)



Input (quantity & quality)

Retina

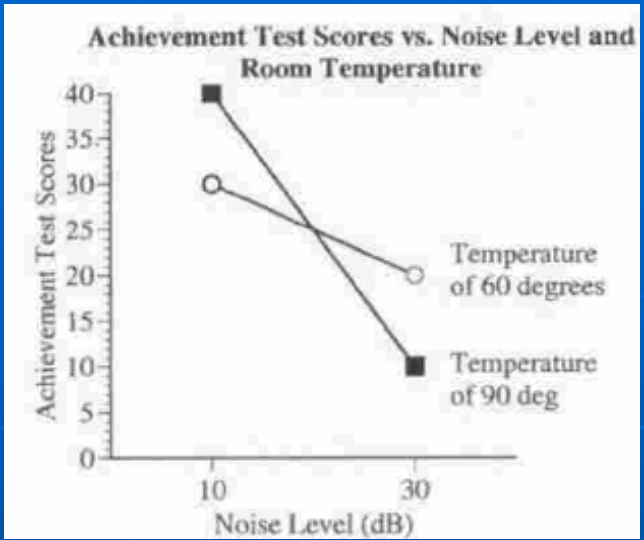
Visual Cortex

Short-term Long-term memory

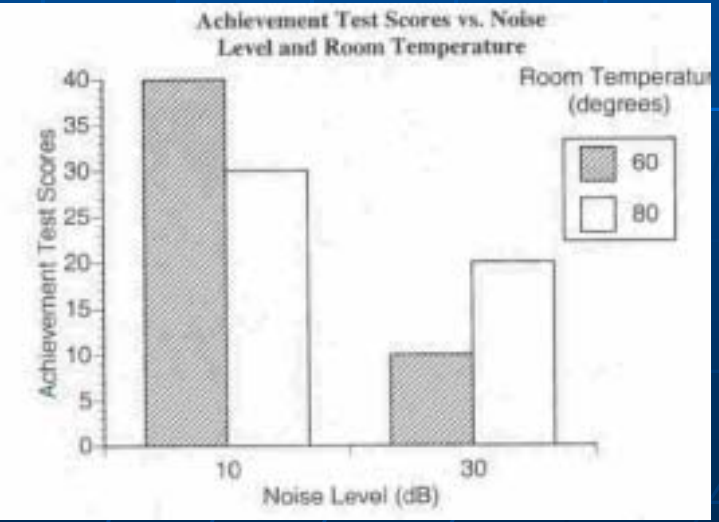
Format

- Line graphs
  - emphasize *x* - *y* trends

  (Carswell et al., 1987; Shan et al., 1999; Zacks & Tversky, 1999)



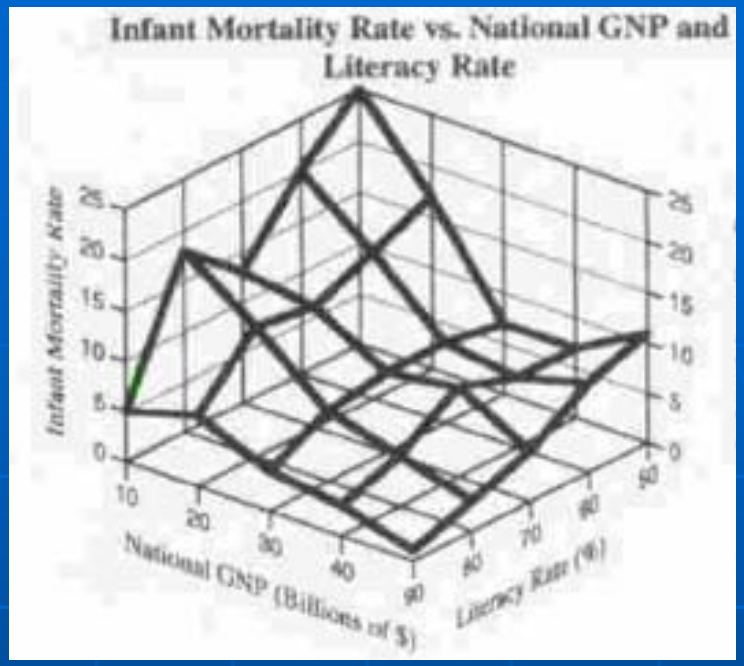
- Bar graphs
  - stimulate discrete comparisons (Shah & Shellhammer, 1999) and
  - reduce bias towards reporting only one variable

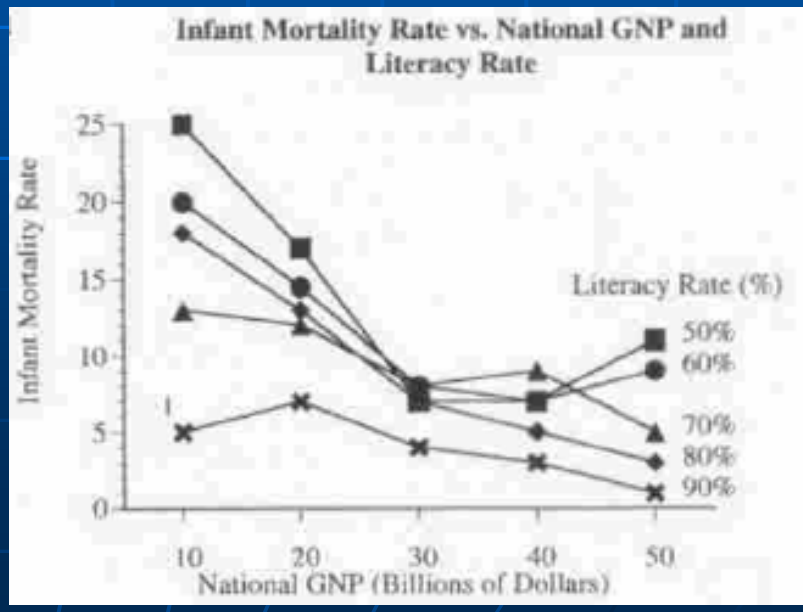


- Pie charts
  - preferable for comparing relative proportions (Kosslyn, 1994; Wilkinson, 1999)

Dimensionality

- 3D graphs enhance comparison of 3 variables (Wickens et al., 1994)
- 3D displays interfere with reading individual data points (Shah et al.)
- 3D objects require prior knowledge of reading similar graphs (Shah et al.)
- For two-variable relationship depicted in 3D, accuracy and speed of data reading decreases/does not change (Fischer, 2000; Spence, 1990)





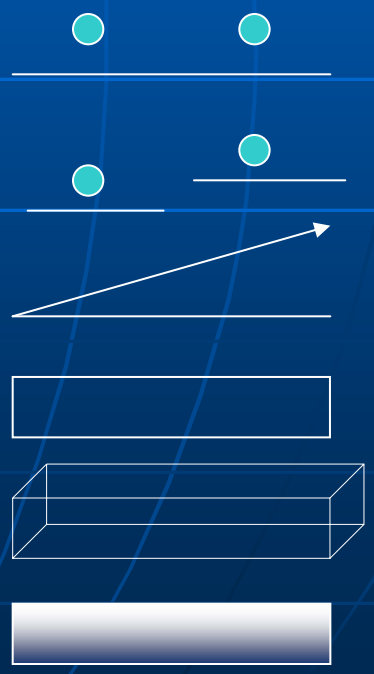
Colors

- Colors may be misleading when used for continuous data (Cleveland & McGill, 1985)
- The use of colors may falsely suggest presence of categorical data (Cleveland & McGill, 1985)
- Colors help to group data in scatter plots (Lewandowsky & Spence, 1989)
- Colors reduce the load of short-term memory when used as referents and are preferred to legends (Kosslyn, 1994)

Rank ordering (most accurate to least accurate) of visual dimensions used to represent quantitative information

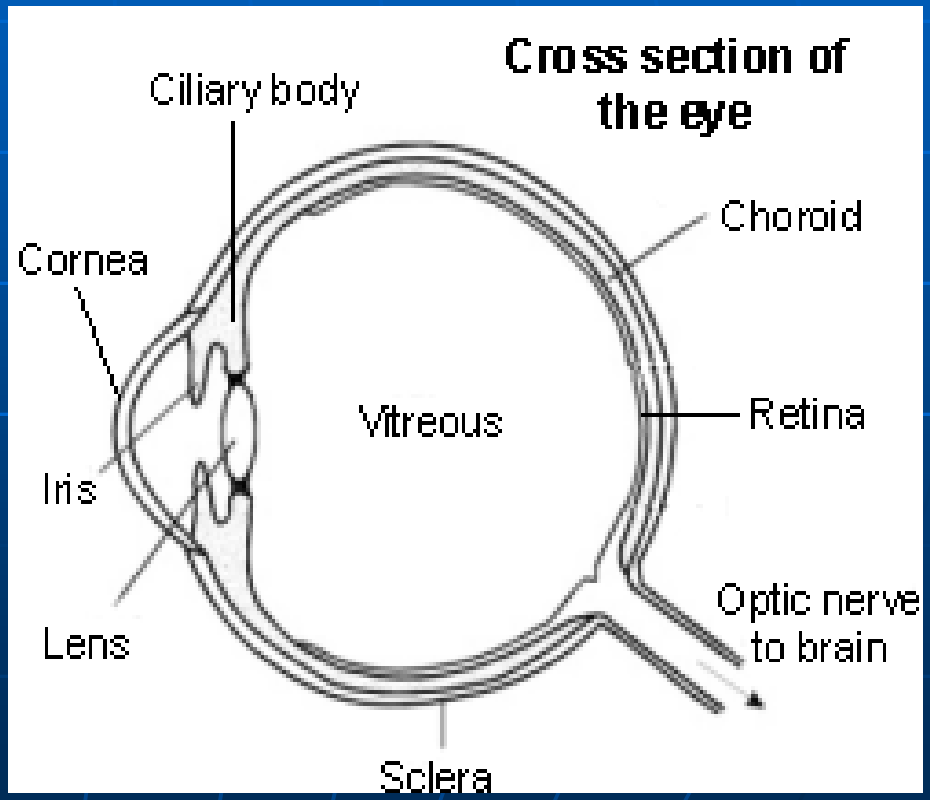
(adopted from Cleveland & McGill, 1984)

1. Position along a common scale
2. Position along nonaligned scales
3. Length, direction, angle
4. Area
5. Volume, curvature
6. Shading, color saturation



The retinal image

- Depends on lightning
- Depends on viewing position
- Depends on color contrasts
- Black & white colors are best suited for visual perception

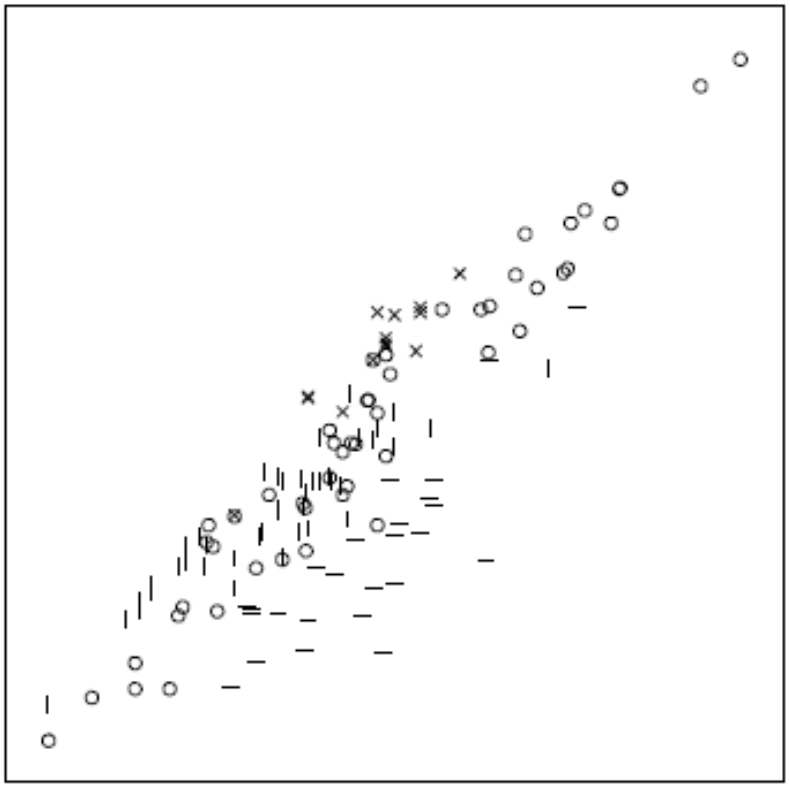


The visual cortex

extracts spatial frequency, orientation, and other features needed to construct complex visual scenes

- Follows the principles of absolute & relative discriminability (Kosslyn, 1989)
- Most discriminable symbols are verticals, horizontals, & diagonals (Cleveland, 1993; Wilkinson, 1993):

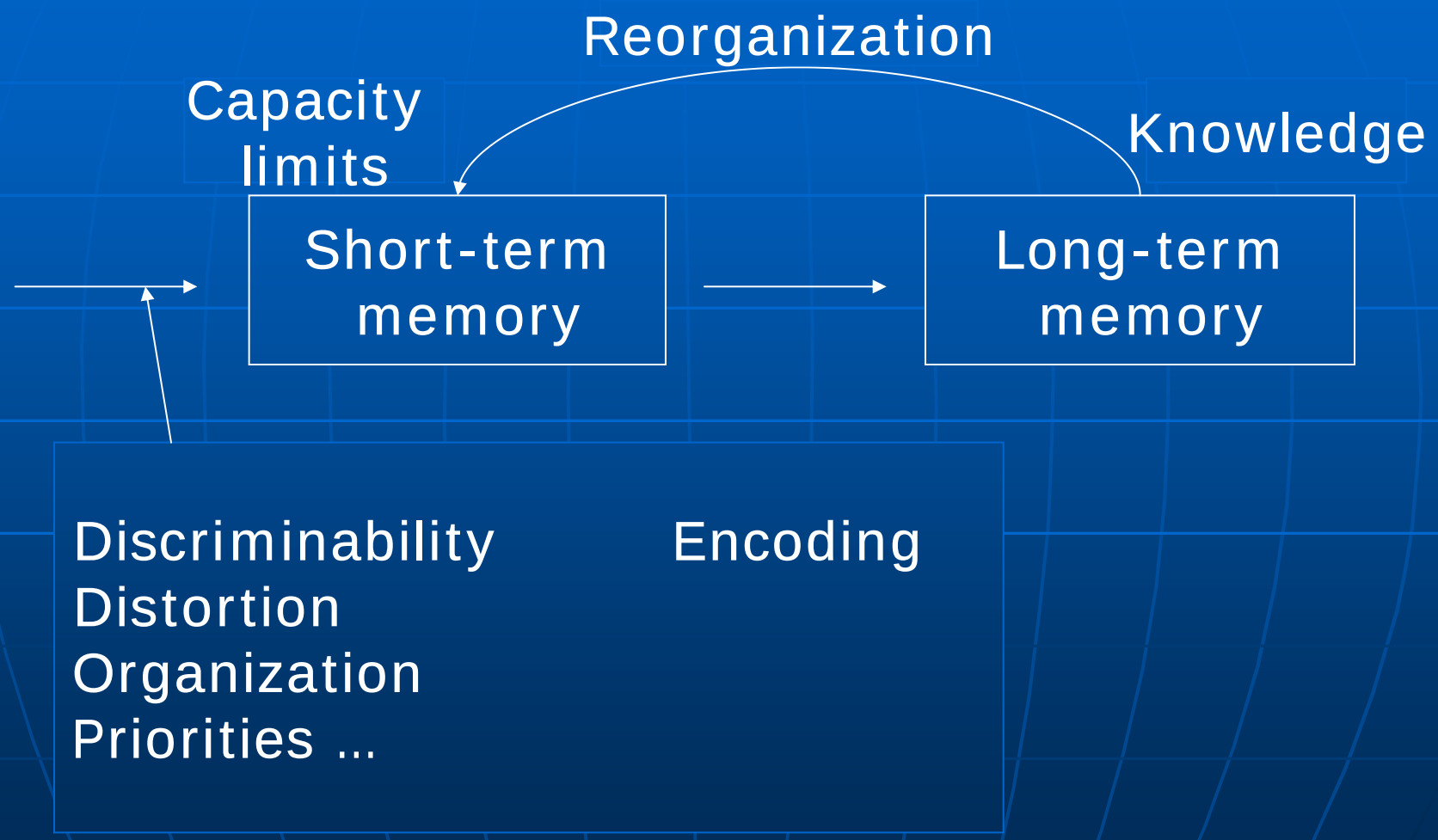
  ‘o’, ‘-’, ‘!’, ‘x’



Principles of perceptual organization

- Gestalt laws:
  - a) good continuity: '----------'
  - b) proximity: 'xx xxx xx'
  - c) similarity: '!!!OOO'
  - d) good form: '{ }' vs. '{ ¥'
- Integral vs. separable dimensions (Kosslyn, 1989)
- Processing priorities (Kosslyn, 1989)
- Processing limitations (Kosslyn, 1989)
  - a) finite capacity (4-7 perceptual groups)
  - b) purposeful unit binding
- Informative mapping (e.g. labels match the axes)

Memory


Reorganization
Capacity limits
Knowledge
Short-term memory
Long-term memory
Discriminability
Distortion
Organization
Priorities ...
Encoding

Knowledge

About graphs

About content

Knowledge is stored in forms of <u>schemas</u> - mental structures that represent some aspect of the world. People use schemas to organize current knowledge and provide a framework for future understanding (Wikipedia)

Knowledge/schemas about graphs that interfere with graph readings:

1. Viewers tends to favor 45 degree lines rather than steeper or flatter lines when required to remember graphical representations of data
   (Schiano & Tversky, 1992; Tversky & Schiano, 1989)

2. Viewers tend to expect independent variable plotted on X axis & dependent variable - on Y axis

3. Viewers confuse slope & height/value magnitude
   (Gattis & Holyoak, 1995 Bell and Janvier, 1981)

4. Novice viewers rely on their prior knowledge about the content rather than the information depicted in the graph (Shah 1995)

Content knowledge

- Influences on interpretation of graphs
- Influences on memory
- Is especially important for novice graph viewers

Easy	Hard
* Distance	* Acceleration
* Position	* Velocity
* Time	

Content knowledge/schemas that interfere with graph readings

- Viewers tend to overestimate the correlation between variables when they believe that these variables are related
  (Freedman and Smith, 1996; Jennings et al., 1982)

- Viewers more likely to make inferences about general trends (main effects) when viewing graphs with familiar relationships
  (Shah & Shellhammer, 1999)

- Younger students tend to interpret abstract representation of data as an iconic representation of real event, e.g. height instead of growth, location instead of speed
  (Janvier 1981; Leinhardt et al., 1990)

? ? ?

Implications

? ? ?

Implications

- Choose the format depending on the communication goal
- Use multiple formats to communicate the same data
- Use 3D with caution
- Choose colors thoughtfully
- Reduce working memory demands
- Follow the principles of perception
- Be aware of your audience

Research proposal

Independent Variables

- Prior knowledge about graphs
- Prior statistical knowledge

Outcomes

- description of relationships b/w 3 variables
- Number of responses
  - Correct
  - Incorrect
- Identification of misleading graphs

Hypotheses

H1: people with prior knowledge of graphs will describe relationship b/w 3 variables

H2: people with prior knowledge of graphs will notice discrepancies in graphical representations

H3: people with prior knowledge of graphs will report more information than people with no such knowledge

H4: people with prior knowledge of graphs will have more correct responses compared to people with no such knowledge

Sample

- Students from the statistical graphing class
- Students from other classes (?)

We will differentiate students by:

- prior knowledge of graphs
- prior statistical knowledge
- gender
- age
- education

Materials

- 4 graphs adopted from other studies (and modified)
- All graphs represent relationship between 3 variables
- Some graphs contain misleading information about the relationship

Procedure

- Students will be given testing instructions
- Students will fill out personal information
- Testing will take 16 minutes (4 minutes per graph)
- Testing materials will be collected and processed

Your feedback

-About hypotheses

-About procedure

-About materials

-???

Selected references

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